

HDOC-13

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits.
- Find the sum of its even digits and the sum of its odd digits separately.

Task1: algorithm:

Step 1: Start.

Step 2: Declare integer variables:

num, temp, digit, choice, even_count, odd_count, even_sum, odd_sum.

Step 3: Initialize:

- even_count = 0
- odd_count = 0
- even_sum = 0
- odd_sum = 0

Step 4: Input a positive integer num.

Step 5: If num <= 0, display "Invalid input! Please enter a positive integer." and stop.

Step 6: Display the menu:

1. Count even and odd digits
2. Find sum of even and odd digits

Step 7: Input the user's choice.

Step 8: Store num in temp.

Step 9: Use switch(choice).

Case 1: Count even and odd digits

1. Repeat while temp > 0.
2. Extract the last digit:
digit = temp % 10
3. If digit % 2 == 0, increment even_count.
4. Otherwise, increment odd_count.

5. Remove the last digit:
temp = temp / 10
6. Display even_count and odd_count.

Case 2: Sum of even and odd digits

1. Repeat while temp > 0.
2. Extract the last digit:
digit = temp % 10
3. If digit % 2 == 0, add it to even_sum.
4. Otherwise, add it to odd_sum.
5. Remove the last digit:
temp = temp / 10
6. Display even_sum and odd_sum.

Default: Display "Invalid choice!"

Step 10: Stop.

Task2: Test case table:

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = 12045, Choice = 1	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	PASS
2	Number = 12345, Choice = 2	Sum_even = 6, Sum_odd = 9	Sum_even = 6, Sum_odd = 9	PASS
3	Number = 2468, Choice = 1	Even_dig = 4, Odd_dig = 0	Even_dig = 4, Odd_dig = 0	PASS
4	Number = 0	Invalid input	Invalid input	PASS

Task3: C program:

```
#include <stdio.h>
int main(void)
{
    int num, temp, digit, choice;
    int even_count = 0, odd_count = 0;
    int even_sum = 0, odd_sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }
    printf("\n1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = num;
    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;
                temp = temp / 10;
            }
            printf("Even_dig = %d\n", even_count);
            printf("Odd_dig = %d\n", odd_count);
            break;
        case 2:
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 == 0)
                    even_sum = even_sum + digit;
                else
                    odd_sum = odd_sum + digit;
                temp = temp / 10;
            }
            printf("Sum_even = %d\n", even_sum);
            printf("Sum_odd = %d\n", odd_sum);
            break;
    }
}
```

```
        default:
            printf("Invalid choice!\n");
        }
    return 0;
}
```

Task4: Output:

```
(kali@kali)-[~/hdcc13.c]
$ ./a.out
Enter a positive integer: 12045

1.Count even and odd digits
2.Find sum of even and odd digits
Enter your choice: 1
Even_dig = 3
Odd_dig = 2
```

```
(kali@kali)-[~/hdcc13.c]
$ ./a.out
Enter a positive integer: 12345

1.Count even and odd digits
2.Find sum of even and odd digits
Enter your choice: 2
Sum_even = 6
Sum_odd = 9
```

```
(kali@kali)-[~/hdcc13.c]
$ ./a.out
Enter a positive integer: 2468

1.Count even and odd digits
2.Find sum of even and odd digits
Enter your choice: 1
Even_dig = 4
Odd_dig = 0
```

```
(kali@kali)-[~/hdcc13.c]
$ ./a.out
Enter a positive integer: 0
Invalid input! Enter a positive integer.
```